

GIT Tutorial

ADVANCED PROGRAMMING

UNIVERSITY OF TEHRAN - COLLEGE OF FARABI

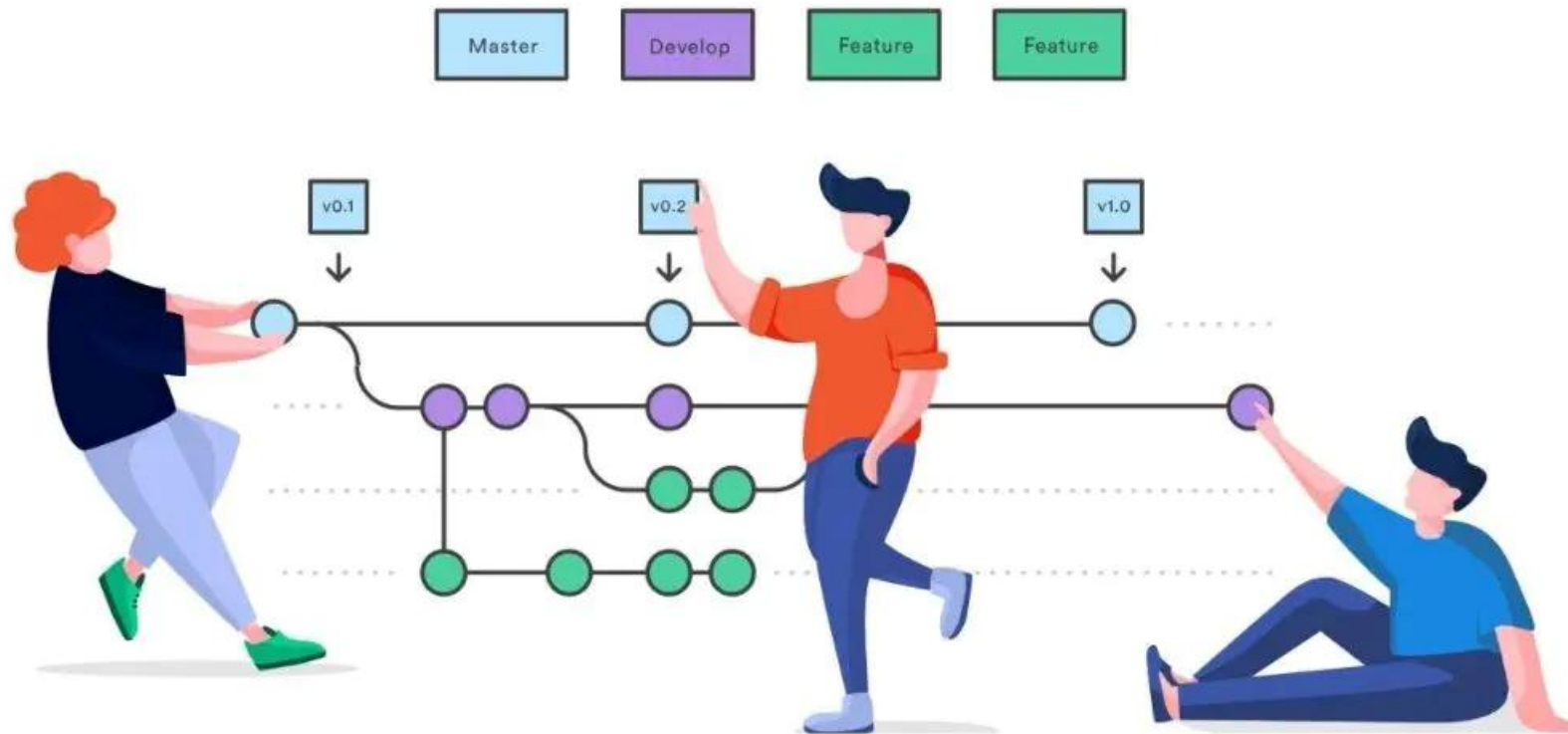
MAY 2026



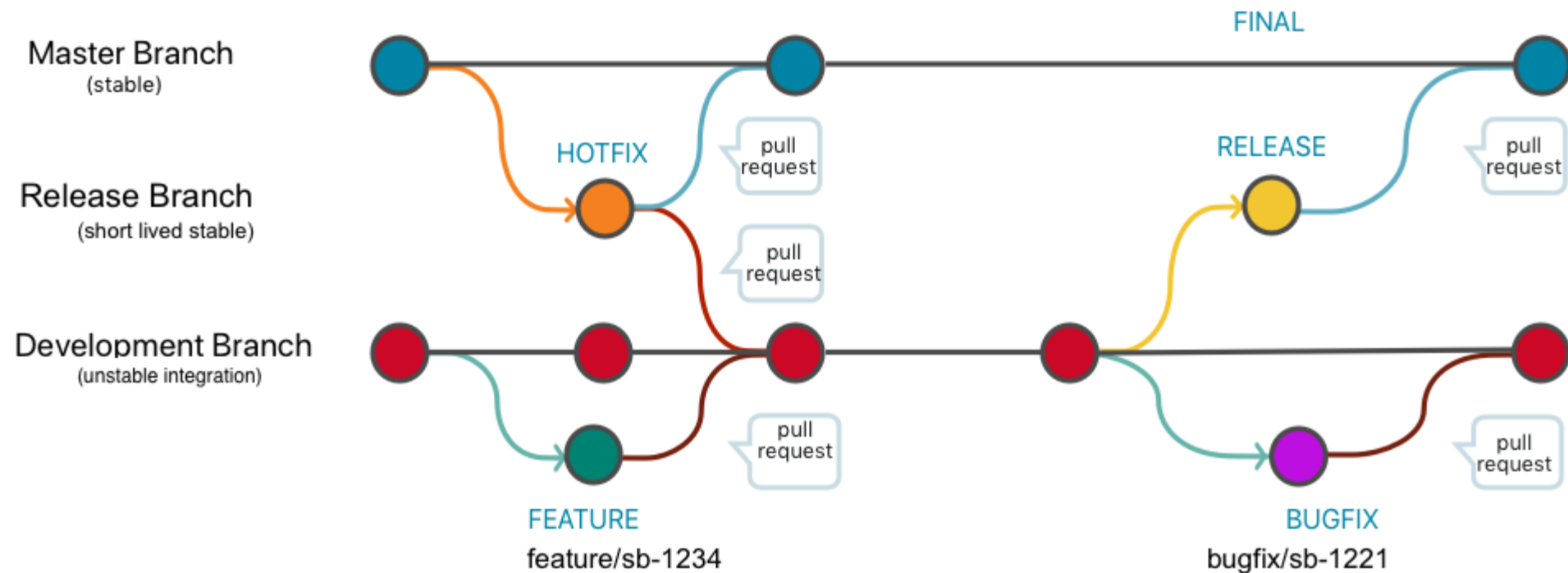
What is git?



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Why do we use git?

- Keep a history of code changes
- Ability to revert to previous versions
- Teamwork without breaking each other's code

Git Essentials

➤ Repository

➤ Staged

➤ Commit

➤ Branch

➤ Merge

How to Define a git Repository?

- Go to the project root.

```
➤ git init
```

Stage files

- `git add .` (to add all changed files)
- `git add <file_path>`

Your First Commit !

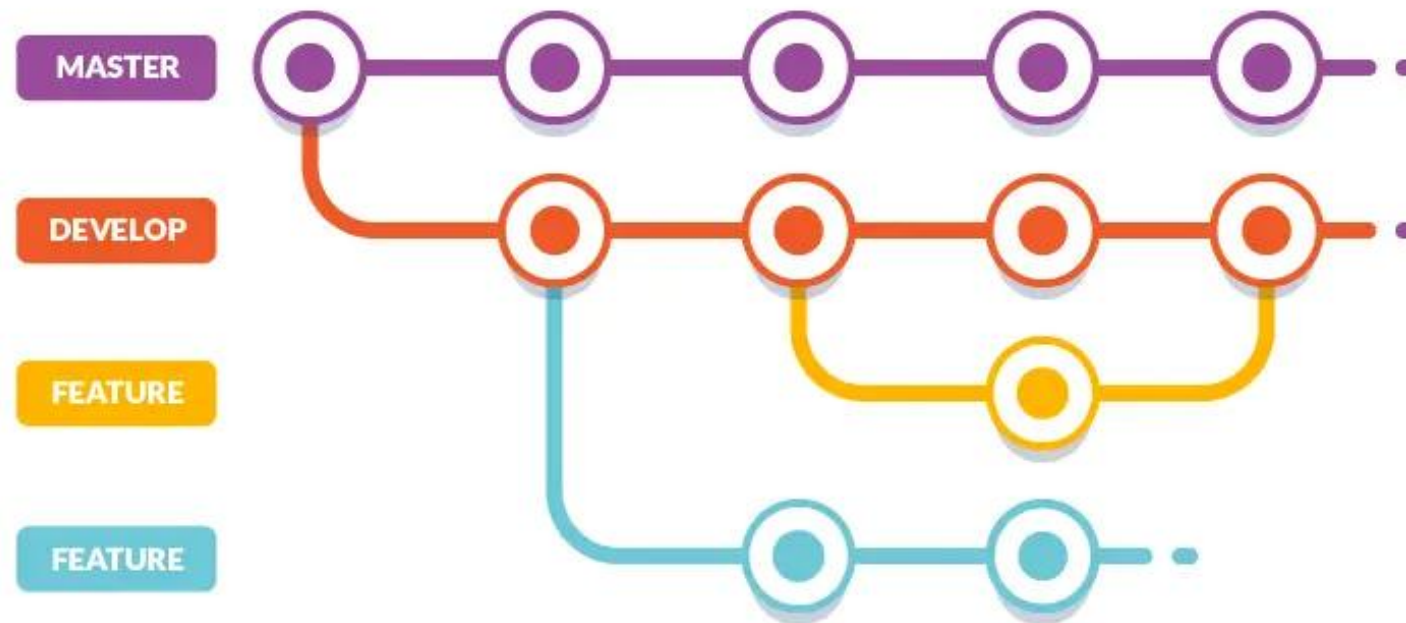
```
➤ git commit -m "your message"
```

Creating and Moving Between Branches

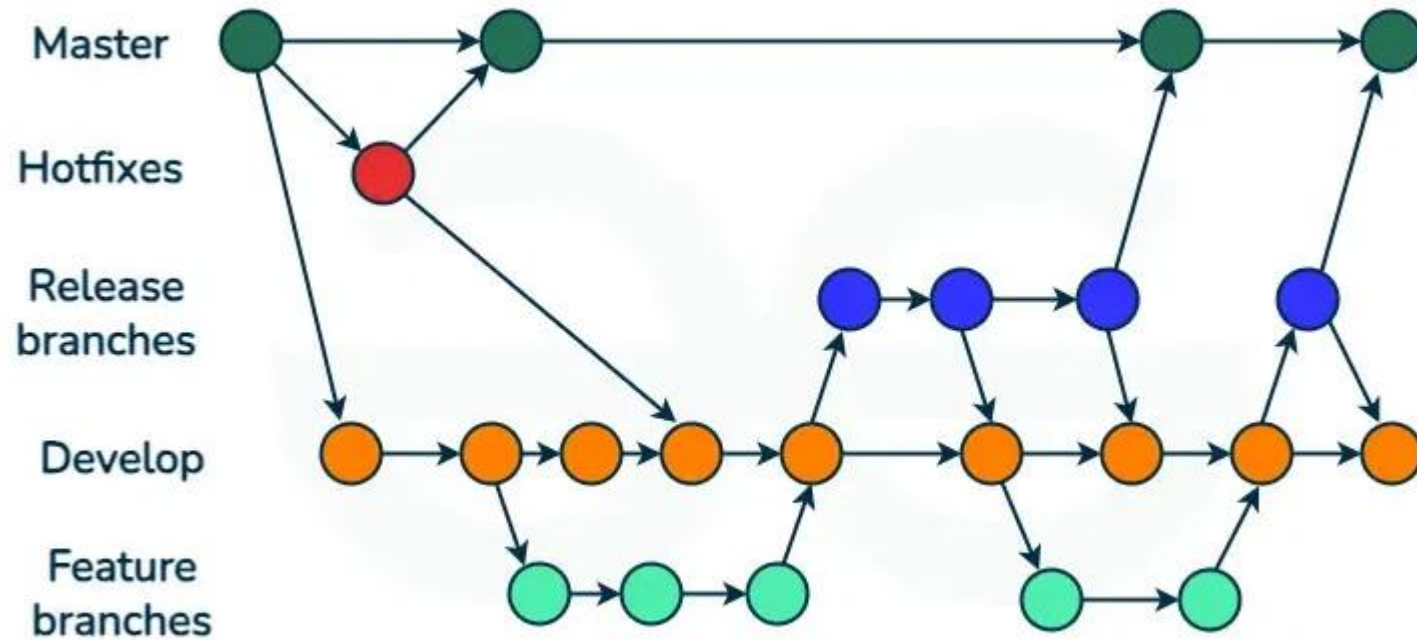
- `git switch -c <your_new_branch_name>` (recommended)
- `git checkout -b <your_new_branch_name>`

- `git switch <your_branch_name>`
- `git checkout <your_branch_name>`

Branches



Branches



Remote

- `git remote add origin <repo_url>`
- `git push origin main (or master)`
- `git pull origin main (or master)`

Git Reset

- `git reset --hard HEAD~<N>`
- `git reset --soft HEAD~<N>`

Git Restore

- `git restore <filename>` (revert the file to the state before adding)
- `git restore --staged` (remove the file from the staging area)

More Advanced

- Conflict
- gitignore
- diff
- Rebase
- Tags
- Stash
- Log
- Revert VS Reset

Thank You!

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